1. Clinical Question & Core Hypothesis

Clinical Question

Triple-negative (ER/PR/HER2-negative) tumors are often TIL-rich, which should predict better outcomes, yet clinical benefit is variable, particularly across patient age — why do abundant lymphocytes sometimes fail to translate into improved survival in TNBC? 2) Tier 1 path(s) used: the single-path chain (Phenotype: triple-negative → Outcome → Age) — I built on the mechanistic tension between TNBC biology, clinical outcome, and age-related modifiers of immunity. I translated 'age' into an immunosenescence/exhaustion mechanism (PDCD1) because age is known to alter T-cell function; I used the TNBC-defining absence of ESR1 as the molecular anchor to keep the focus on triple-negative tumors. 3) Tier 2 supporting evidence: (a) The dataset includes T-cell exhaustion markers (PDCD1, CTLA4, LAG3, TIGIT) that plausibly mediate functional impairment of TILs; (b) published evidence links immunosenescence and increased checkpoint expression with reduced anti-tumor immunity in older hosts; (c) TNBC is a context where TIL density is prognostic but heterogeneously so, supporting a modifier hypothesis. 4) Why OS as endpoint: OS is the most robust, hard clinical endpoint available in TCGA-BRCA and best captures the net effect of tumor biology, age, and immune competence in a heterogeneous cohort. 5) Deliberate exclusions: I excluded modeling/architecture nodes (DenseNet, ensemble, DeepProg) and therapeutic-repurposing nodes (imatinib, BCR–ABL) because the current puzzle is mechanistic (why TILs fail in TNBC with age) rather than a model-performance or drug-repurposing question. I also excluded other exhaustion/proliferation genes from the core hypothesis to keep the mechanistic claim sharp; they appear as secondary_modifiers for follow-up analyses?

Working Hypothesis

In ESR1-negative (triple-negative) breast cancers, high tumor-infiltrating lymphocyte (TIL) density is associated with improved overall survival (OS) in younger patients but not in older patients, due to age-related T-cell exhaustion driven by upregulation of PDCD1 (PD-1) and co-inhibitory receptors (CTLA-4, LAG-3, TIGIT) on TILs. This leads to a three-way interaction where the protective effect of TILs on OS is significant in younger patients with lower exhaustion marker expression, attenuated or reversed in older patients with higher exhaustion marker expression. Secondary modifiers (CTLA-4, LAG-3, TIGIT, MKI67) will refine the exhaustion/activation status but the primary mechanistic claim centers on PD-1–mediated exhaustion modulating TIL efficacy in ESR1-negative tumors. The hypothesis predicts that the age-dependent TIL-OS association identifies subgroups that may benefit from therapies targeting PD-1 or combination checkpoint blockade, particularly in older TNBC patients with high TIL density and an exhaustion signature.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	PDCD1, ESR1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PDCD1: PDCD1, CTLA4, LAG3, TIGIT) that plausibly mediate functional impairment of TILs; (b) published evidence links immunosenescence and increased checkpoint expression with reduced anti-tumor immunity in older hosts; (c) TNBC is a context where TIL density is prognostic but heterogeneously so, supporting a modifier hypothesis.

ESR1: ESR1 as the molecular anchor to keep the focus on triple-negative tumors.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTLA4	Supporting modifier from knowledge graph
Secondary	LAG3	Supporting modifier from knowledge graph
Secondary	TIGIT	Supporting modifier from knowledge graph
Secondary	MKI67	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Assemble ESR1-negative breast cancer cohorts with available OS data, TIL density assessments (pathology or gene-expression–based proxies), and expression data for PDCD1 and co-inhibitory receptors (CTLA-4, LAG-3, TIGIT), plus MKI67 as a proliferation marker.

Prediction 2: Stratify by age (e.g., <50 vs ≥50) and by exhaustion-marker expression (low vs high).

Prediction 3: Use Cox proportional hazards models to test for a three-way interaction: TIL_density × age_group × exhaustion_signature (PDCD1 primary; CTLA-4/LAG-3/TIGIT as modifiers).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTLA4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAG3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.